

Eric Bolton

Machine Learning Engineer & Computational Journalist

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EXPERIENCE

Spherical Harmonics | Co-founder

Present

- Co-founded a nonprofit building open-source tools that help communities design safer streets.

Meta | Software Engineer, Instagram Reels

Apr 2025 - May 2026

- Co-led development of a distilled multimodal model for time-sensitive Reels, improving precision from 0.752 to 0.889 at 80% recall; owned training data, evaluation, feature registration, and experiments.
- Built trend-retrieval and ranking infrastructure, expanding candidate media pools and implementing feature-based filtering to improve serving capacity without reducing recall.
- Created reusable evaluation pipelines for timeliness, trend expansion, and clustering, with representative sampling, automated LLM scoring, and dashboards for comparing model quality.

NewsWyze | Chief Technology Officer

Feb 2024 - Apr 2025

- Led development of production LLM summarization and translation systems for newsrooms, helping reduce editorial processing time by 85% while preserving editorial quality across languages.
- Built retrieval-augmented generation pipelines using journalistic annotations to improve model outputs, integrating evaluation and monitoring into the production workflow.

Applied XL | Founding Engineer

Jun 2021 - Feb 2024

- Managed two data scientists building data-to-text LLMs that surfaced investment opportunities for biotech funds; used LoRA and DPO fine-tuning and evaluated outputs with BERTScore and FActScore.
- Served as interim CTO, overseeing the Django API, React frontend, data workflows, and AWS infrastructure; developed deterministic graph clustering and production classifiers for drug-pipeline events.
- Co-authored a clinical-trials report analyzing early stopping across 70,000 trials, using Weibull regression to investigate risk factors and patterns in trial outcomes.

Columbia University, Brown Institute | ML Engineer, Local News Lab

Jun 2020 - Jun 2021

- Built article recommendation systems using matrix factorization and document embeddings, balancing recency, relevance, and diversity for local news audiences.
- Prototyped adaptive paywalls and alternative revenue models, and ran A/B tests and contextual-bandit experiments to evaluate recommendation strategies.

The Wall Street Journal | Machine Learning Scientist, R&D

Jun 2018 - Jun 2020

- Built Talk2020's NLP pipeline and search tools for decades of political statements, making candidate quotations searchable for journalists and millions of subscribers.
- Developed AI Portraits, a GAN-based system that turns photographs into WSJ-style stipple drawings; augmented training data to improve representation across portrait subjects.
- Analyzed a corpus of more than one million articles with topic models and word embeddings, informing product strategy and reporting on changes in news coverage over time.

SELECTED RESEARCH & PROJECTS

High Quality Real-Time Structured Debate Generation: Co-authored research using debate-tree data augmentation to improve language-model argument generation.

Frontier Beat: Built a platform tracking and summarizing trending arXiv papers and associated GitHub repositories.

EDUCATION

Columbia University | M.S. Computer Science; M.S. Journalism.

May 2019

Brown Center Fellow; Machine Learning Teaching Assistant.

Princeton University | A.B. Geosciences.

May 2015

SKILLS & LANGUAGES

Languages: Python, SQL, Hack/PHP, Java, C, Go, R

ML & infrastructure: PyTorch, JAX, TensorFlow, Transformers, PEFT, TRL, XGBoost, PySpark, AWS, Terraform, Airflow

Spoken languages: English and French (native); Spanish (intermediate)